Exp 2a – Atomic Mass

INTRODUCTION

In this experiment, you will be calculating the average atomic mass of a fictional element called "Fresnocitium" which has three isotopes that are different colors: red, black, and white.

BACKGROUND

The "atomic mass" reported on periodic tables is the average atomic mass of all the different isotopes of the given element, weighted by the natural abundance. Atoms of real elements are too small for us to see with our eyeballs, or touch individually to get a sense of what it means to have a weighted average mass. Isotopes that are more abundant count towards the average atomic mass more than less abundant isotopes.

Atoms of fresnocitium are large enough to be picked up and counted with your fingers, allowing us to determine the abundance of each isotope and calculate the average atomic mass of fresnocitium with only an analytical balance. The abundance of an isotope can be calculated by:

$$Abundance\ (\%) = \frac{number\ of\ atoms}{total\ number\ of\ atoms} \times 100$$

The average atomic mass is calculated using the equation below, with the abundances in decimal form. To convert from the percent abundance into decimal form (fractional abundance), divide the percentage by 100.

$$Average\ Atomic\ Mass = (mass_1 \cdot abundance_1) + (mass_2 \cdot abundance_2) \ldots$$

ADDITIONAL READING

- Atomic Mass

SUPPLY LIST

- Beaker (100 mL)
- Tray or paper plate

CLASSROOM SUPPLIES

These are shared supplies or reactants. Do not gather these items. Leave these items at their designated location in the classroom and use as needed according to the procedure.

- Beads made of various materials

PROCEDURE

1. ☐ Bring an empty 100 mL beaker over to the analytical balances.
2. ☐ Tare (zero) the balance so that the display reads 0.0000g. Place the empty beaker on the balance and then record the mass on your report sheet. You will repeat these instructions twice: Trial 1 & Trial 2. You will use this same beaker to record all the mass measurements for both trials.
3. ☐ Shake the supply bin of Fresnocitium thoroughly. Ensure the lid is snapped on and hold the lid on while shaking and flipping the bin over.
4. ☐ Use the beaker as a scoop to obtain a sample of Fresnocitium atoms (beads). Fill the beaker with beads.
5. ☐ Return to the balances. Tare (zero) the balance and then record the mass of the beaker containing Fresnocitium.
6. ☐ Carefully sort the different isotopes of fresnocitium (Red, Black, and White). You may obtain other beakers to hold these atoms if you like, as long as you do not use them for weighing.
7. ☐ Place all the atoms of the red isotope in your beaker.
8. ☐ Tare the balance and then record the mass of the beaker with the red isotope.
9. ☐ Remove the red atoms from the beaker. Count and record the number of red atoms. You should double-check that your counting is accurate.
10. ☐ Repeat steps 6-9 with the black atoms and then white atoms.
11. ☐ Return your atoms to the supply bin.
12. ☐ Using the same 100 mL beaker, repeat steps 3-10 to obtain a second trial of data.

CLEAN-UP

1. ☐ After your data is approved by your instructor, return all the atoms to the supply bin. With lid securely on the bin, shake the bin thoroughly.

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

DATA

1. Mass of beaker (empty): _____

	Trial 1	**Trial 2**
Mass of beaker + "Nature's Mix" of Fresnocitium		
Mass of beaker + Red Atoms		
Mass of beaker + Black Atoms		
Mass of beaker + White Atoms		
Number of Red Atoms		
Number of Black Atoms		
Number of White Atoms		

CALCULATIONS

Show your work neatly, with units, where applicable.

2. A. Mass of "Nature's Mix" of Fresnocitium:

TRIAL 1	**TRIAL 2**

B. Total Number of Atoms:

TRIAL 1	**TRIAL 2**

C. Average Mass per Atom in Nature's Mix:

TRIAL 1	**TRIAL 2**

3. A. Complete the table below for Trial 1:

	TRIAL 1	
Isotope Color	Percent Abundance of Isotope (%):	Average Mass per atom of isotope (g):
Red		
Black		
White		

Trial 1: Average Atomic Mass based on equation:

_____

B. Complete the table below for Trial 2:

	TRIAL 2	
Isotope Color	Percent Abundance of Isotope (%):	Average Mass per atom of isotope (g):
Red		
Black		
White		

Trial 2: Average Atomic Mass based on equation:

_____

4. Compare the "average mass per atom" obtained in Question 2C to the "average atomic mass" obtained in Question 3A for Trial 1. Do these values match? Should they match?
   Write in complete sentences and include both numerical values obtained.

5. Are your results from Trial 1 similar to the results found in Trial 2 for Question 3? If not, why not? (Refer back to the tables in Question 3 to determine what was different between trials)
   Write in complete sentences and include numerical values obtained.

6. If you used a much larger sample of fresnocitium, would it make a substantial difference to your calculated average atomic mass? Why or why not?
   Write in complete sentences.